

MPS-3206

Economic Regulated Power Supply User Manual

IV Panel feature

1. Front pannel

I Overview

MPS-3206 Series is an economical regulated DC power supply that offers reliable performance and high stability. It features a 4-digit LED voltage/current display and includes overvoltage protection (OVP), overcurrent protection (OCP), and overtemperature protection (OTP). OVP and OCP setpoints are user-adjustable. It is ideal for college teaching labs, factory production lines, and equipment maintenance.

II Specification

Model	Voltage	Current	LED Display
MPS-3206	0-32V	0-6.1A	4 bits
MPS-3210	0-32V	0-10A	4 bits
MPS-6205	0-62V	0-5.1A	4 bits

III Technical Specifications

1. Rated Operating Conditions

Input voltage: AC 220 V $\pm$ 10%, 50 Hz

Operating conditions: Temperature: 0 – 40°C; Relative humidity: $\leq$ 80% RH

Storage conditions: Temperature: -15 – 60°C; Relative humidity: $\leq$ 80% RH

2. Test conditions: 23°C $\pm$ 5°C

Model		MPS-3206	MPS-3210	MPS-6205
Rated input voltage		AC220V/110V $\pm$ 10%		
Rated output voltage		0~32V	0~32V	0~62V
Rated output current		0~6A	0~10A	0~5A
Load regulation	Voltage	$< 0.1\% + 5\text{mV}$	$< 0.1\% + 10\text{mV}$	$< 0.1\% + 5\text{mV}$
	Current	$< 0.2\% + 3\text{mA}$	$< 0.2\% + 5\text{mA}$	$< 0.2\% + 3\text{mA}$
Line regulation	Voltage	$< 0.01\% + 5\text{mV}$	$< 0.01\% + 10\text{mV}$	$< 0.01\% + 5\text{mV}$
	Current	$< 0.2\% + 3\text{mA}$	$< 0.2\% + 5\text{mA}$	$< 0.2\% + 3\text{mA}$
Setting accuracy	Voltage	$\leq 0.1\% + 1 \text{ digit}$	$\leq 0.1\% + 1 \text{ digit}$	$\leq 0.1\% + 2 \text{ digits}$
	Current	$\leq 0.2\% + 3\text{mA}$	$\leq 0.2\% + 5\text{mA}$	$\leq 0.2\% + 3\text{mA}$
Setting resolution	Voltage	10mV	10mV	10mV
	Current	1mA	1mA	1mA
Readback accuracy	Voltage	$\leq 0.1\% + 1 \text{ digit}$	$\leq 0.1\% + 1 \text{ digit}$	$\leq 0.1\% + 2 \text{ digit}$
	Current	$\leq 0.2\% + 3\text{mA}$	$\leq 0.2\% + 5\text{mA}$	$\leq 0.2\% + 3\text{mA}$
Readback resolution	Voltage	10mV	10mV	10mV
	Current	1mA	1mA	1mA
Ripple	Voltage	$\leq 10\text{mVrms}$	$\leq 10\text{mVrms}$	$\leq 10\text{mVrms}$
	Current	$\leq 5\text{mA}_{\text{rms}}$	$\leq 5\text{mA}_{\text{rms}}$	$\leq 5\text{mA}_{\text{rms}}$
OVP		0~32V $\pm$ 0.2%FS	0~32V $\pm$ 0.2%FS	0~62V $\pm$ 0.2%FS
Max. voltage		32V $\pm$ 0.2%	32V $\pm$ 0.2%	62V $\pm$ 0.2%
OCP		0~6.1A $\pm$ 0.2%FS	0~10A $\pm$ 0.2%FS	0~5.1A $\pm$ 0.2%FS
Max. current		6.1A $\pm$ 0.2%	10A $\pm$ 0.2%	5.1A $\pm$ 0.2%
Temperature	Operation	0 ~ 40°C $\leq$ 80%RH		
Coefficient	Storage	-15~70°C $\leq$ 80%RH		
Size (W*H*D)	mm	115*96*261		
Packing size (W*H*D)	mm	167*153*317		
Net weight	kg	1.5		
Gross weight	kg	1.9		



1. Voltage display window — Displays the output voltage
2. Current display window — Displays the output current
3. M1 – M5 save/recall key — Short press to recall a stored setting, the indicator lights up. Press and hold to save the current setting. After setting voltage and current, press and hold the key until the indicator lights up; the setting is then saved.
4. POWER switch — Turns the instrument power on/off
5. Output voltage/current adjustment knob — Used with the V/I setting key (item 7). In standby or output mode, press V/I. The voltage display flashes. Turn the knob clockwise to increase the output voltage or counterclockwise to decrease it. Press the knob to select the flashing digit. To set current, press V/I to switch to the current display; the current display flashes. Turn clockwise to increase output current or counterclockwise to decrease it. Press the knob to select the flashing digit.
6. OVP/OCP setting key — Press and hold OVP/OCP to enter OVP setting mode.
When the display shows “OVP,” rotate the knob to change the setpoint (same method as voltage/current setting). In OVP setting mode, short-press OVP/OCP to enable or disable OVP. When enabled, “OVP” appears in the upper-right corner of the display.

Press and hold OVP/OCP again to enter OCP setting mode. When the display shows “OCP,” rotate the knob to change the setpoint (same method as voltage/current setting). In OCP setting mode, short-press OVP/OCP to enable or disable OCP. When enabled, “OCP” appears in the upper-right corner of the display.

Press and hold OVP/OCP again to return to OVP setting mode. If no setting is made, the display will stop flashing after a few seconds and exit setting mode. When OVP/OCP is enabled, protection is triggered if the actual output exceeds the setpoint. If the display shows “EROCP” or “EROVP,” press ON/OFF to clear it.
7. Voltage/current setting key — Used with the adjustment knob (see item 5).
8. ON/OFF key — Controls the output on/off. When output is on, the ON/OFF indicator in the lower-right corner lights up; when off, it is dark. If the instrument is in OVP/OCP protection, press this key to reset it.

9. Output terminal negative (black).
10. Output grounding terminal (green) — Connected to the chassis and to the grounding wire of the power cord.
11. Output terminal, positive (red)
12. CV indicator (constant voltage) — Lights when the instrument is operating in constant-voltage mode.
13. CC indicator (constant current) — Lights when the instrument is operating in constant-current mode.

2 . Rear panel



V Operating Instructions and Protection Measures

1. Set the POWER switch to OFF. Plug in the power cord. Make sure the L terminal of the power cord is connected to the hot (live) wire of the power outlet, and that the ground wire is properly connected to earth ground.
2. The MPS-3206 power supply must be grounded.
3. Ensure adequate ventilation during use. Maintain at least 10 cm (4 in.) of clearance above, below, left, and right of the unit to allow proper airflow. Do not use this power supply in dusty or corrosive environments or near hazardous substances.
4. Technical specifications should be measured after the power supply has been powered on for 15 minutes.
5. Before power-on, check that the input voltage selector switch matches the local AC line voltage. Otherwise, the unit may be seriously damaged.

VI Packing List

1. MPS-3206 × 1 pc
2. Power cord × 1 pc
3. Test lead × 1 pair
4. User manual × 1 pc

VII Maintenance

1. If the AC input voltage is normal but the digital display does not light up after power-on, the fuse may be blown or there may be another fault. Turn off the power switch, unplug the power cord, replace the fuse, or have a qualified technician inspect the unit.
2. During operation, if the output current exceeds the preset value and the CC indicator lights, the power supply is in current limiting / constant-current (CC) mode. The unit automatically switches to constant-current operation. Check the load, or increase the current setting as required.
3. During operation, if the output current is below the preset value and the CV indicator lights, the power supply is in constant-voltage (CV) mode. The unit automatically switches to stable voltage operation. Check the load, or increase the voltage setting as required.
4. If the problem cannot be resolved, contact your local distributor or our company.

Warranty Card

What This Warranty Covers:

If the product fails due to a defect in materials or workmanship, MATRIX will provide free repair service during the warranty period. If the product fails due to improper operation, misuse, or negligence, MATRIX will provide repair service at the customer ' s expense, even within the warranty period.

How Long This Warranty Lasts:

This warranty is valid for three (3) years from the date of original purchase for all MATRIX-branded products.

Who Is Covered:

This warranty covers only the original purchaser of this product. It is not transferable to subsequent owners or purchasers.

How to Obtain Warranty Service?

If your product develops a problem, please contact your local distributor first. If you cannot locate a local distributor, you may contact us directly: **Email: service@szmatrix.com, Phone: +86 755 2836 4276**

What information do customers need to supply?

Model No.	
Serial No.	
Problem description	
Picture	
Video if necessary	